RESEARCH NOTE, ROUND 2 · FOLLOWS “MERSENNE DIVISOR DOUBLING”

$f(2n)/f(n)$ for $f(n)=\sum_{k\le n}\tau(2^k-1)$: new theorems and the exact barrier

Nine independent lines of attack, one adversarial audit, and a certified factor table for $k \le 300$.

NOT DECIDED	Neither $f(2n)/f(n)\to\infty$ nor $\liminf f(2n)/f(n)<\infty$ is proved. Every route stops at the same statement about individual numbers 2^p-1 (Section 6), and Section 5 shows why: models that satisfy every soft property of 2^k-1 used anywhere in this work give either answer.
PROVED	There is **no finite constant** c with $f(2n)/f(n)\to c$: $\limsup=\infty$ (Theorem 2).
PROVED	$f(2n)\ge 3f(n)$ for every $n\ge 1$, with equality only at $n=1,2,4$ (Theorem 3). If the Fermat number F_a is composite, $f(2n)\ge\left(4-\frac{16}{3\cdot 2^{a-2}+4}\right)f(n)$ on $[2^a,2^{a+1})$ (Theorem 4).
PROVED	For every C, the set $\{n : f(2n)>Cf(n)\}$ meets every chain $n,2n,\dots,2^{J(C)-1}n$ and has lower density $\ge 1/(2^{J(C)}-1)$ (Theorem 5).
PROVED	Exact identity $f(2n)-2f(n)=\#\{d \text{ odd} : \{n/\mathrm{ord}_d 2\}\ge\frac12\}$ (Theorem 1), and an exact reduction of the question to “thin” indices (Theorem 6).

1 An exact identity

Write $T(k)=\tau(2^k-1)$ and $o(d)=\mathrm{ord}_d(2)$ for odd d. Every divisor of 2^k-1 is odd, and an odd d divides 2^k-1 iff $o(d)\mid k$.

Theorem 1. $f(n) = \sum_{d \text{ odd}} \lfloor n/o(d) \rfloor$, *hence* $f(2n) - 2f(n) = \#\{d \text{ odd} : \{n/o(d)\} \geq \frac{1}{2}\}$, *where* $\{\cdot\}$ *is the fractional part.*

Proof. $f(n) = \sum_{k \leq n} \#\{d : o(d) \mid k\} = \sum_d \#\{k \leq n : o(d) \mid k\}$. For real x, $\lfloor 2x \rfloor - 2\lfloor x \rfloor$ is 1 if $\{x\} \geq \frac{1}{2}$ and 0 otherwise; apply with $x = n/o(d)$. ∎

So the question is whether, for every n, the odd moduli d whose order $o(d)$ puts n in the upper half of a period outnumber $\sum_d \lfloor n/o(d) \rfloor$ by an unbounded factor.

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2 No finite limit

Theorem 2. $\limsup_n f(2n)/f(n) = \infty$. *In particular there is no finite* c *with* $f(2n)/f(n) \to c$.

Proof. If $f(2n) \leq Cf(n)$ for all $n \geq n_0$, then $f(x) \leq C' x^{\log_2 C}$. But for the primorial $P \leq n$ with $\tau(P) = 2^{(1+o(1)) \log n / \log\log n}$, Zsigmondy's theorem gives a distinct primitive prime for every divisor $d \neq 1, 6$ of P, so $f(n) \geq T(P) \geq 2^{\tau(P)-2}$, which is not polynomial in n. (This is Theorem 8 of the earlier note.) ∎

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3 The sharp constant 3

Notation: $F_i = 2^{2^i} + 1$, so $T(2^a) = \prod_{i<a} \tau(F_i)$ and $T(2^{a+1}) = \tau(F_a)\, T(2^a)$. In general $T(2k) = T(k)\, \tau(2^k + 1)$ because $2^k - 1$ and $2^k + 1$ are coprime.

Lemma 3.1. $\tau(2^m + 1) \geq 4$ *unless* m *is a power of 2 or* $m = 3$*; and* $\tau(9) = 3$.

Proof. If m has an odd divisor $u > 1$, then $2^{m/u} + 1$ is a proper divisor, so $2^m + 1$ is composite. It has exactly three divisors only if $2^m + 1 = q^2$, i.e. $(q-1)(q+1) = 2^m$, forcing $q = 3$, $m = 3$. ∎

Lemma 3.2. *Let* $r = 2^b u$ *with* $u > 1$ *odd and* $C(r) = \sum_{i=0}^{b} T(2^i u)$. *Then* $C(r) \leq T(2r)/3$.

Proof. $T(2^{i+1}u) = T(2^i u)\, \tau(2^{2^i u} + 1)$, and by Lemma 3.1 each factor is ≥ 4, except the factor $\tau(9) = 3$ when $i = 0$, $u = 3$. If all factors are ≥ 4, $C(r) \leq T(2r) \sum_{j \geq 1} 4^{-j} = T(2r)/3$. If $u = 3$, $C(r) \leq T(2r)\left(\sum_{j=1}^{b} 4^{-j} + \frac{1}{3 \cdot 4^b}\right) = T(2r)/3$. ∎

Lemma 3.3. *For* $c \geq 1$*:* $T(3 \cdot 2^c) \geq \frac{3}{2}\, 2^c\, T(2^c)$.

Proof. The divisors of $3 \cdot 2^c$ that do not divide 2^c are $3 \cdot 2^i$, $0 \leq i \leq c$. For $i \neq 1$ each has a

Proof. The divisors of $3 \cdot 2^c$ that do not divide 2^c are $3 \cdot 2^i$, $0 \le i \le c$. For $i \ne 1$ each has a primitive prime (Zsigmondy; the only exception is 6), these are distinct, differ from 3, and do not divide $2^{2^c} - 1$: a factor 2^c. By lifting the exponent, $v_3(2^{2^c} - 1) = 1$ and $v_3(2^{3 \cdot 2^c} - 1) = 2$, turning the factor 2 contributed by the prime 3 into 3: a further factor $3/2$. ∎

Theorem 3. *$f(2n) \ge 3f(n)$ for every $n \ge 1$, with equality exactly for $n \in \{1, 2, 4\}$.*

Proof. For $n \le 15$ this is a direct computation (ratios $3, 3, 3.4, 3, 3.55, \dots, 5.94$). Let $n \ge 16$, $2^a \le n < 2^{a+1}$, $a \ge 4$, and $R = \mathbb{Z} \cap (n/2, n]$. Each $m \le n$ is uniquely $r/2^j$ with $r \in R$, $0 \le j \le v_2(r)$, so $f(n) = \sum_{r \in R} C(r)$ with $C(r) = \sum_j T(r/2^j)$; the numbers $2r$ are distinct elements of $(n, 2n]$, so

$$f(2n) - 3f(n) \;\ge\; \sum_{r \in R} \big(T(2r) - 2C(r) \big).$$

For $r \ne 2^a$, Lemma 3.2 gives a term $\ge T(2r)/3 \ge 0$. For $r = 2^a$: since $T(2^{i+1}) \ge 2T(2^i)$, $\sum_{i<a} T(2^i) < 2T(2^{a-1})$, so that term is $> (\tau(F_a) - 2)T(2^a) - 4T(2^{a-1}) \ge -4T(2^{a-1})$. Compensate with $r_0 = 3 \cdot 2^{a-2}$ if $n < 3 \cdot 2^{a-1}$ and $r_0 = 3 \cdot 2^{a-1}$ otherwise; in both cases $r_0 \in R$ and, by Lemma 3.3, $T(2r_0)/3 \ge T(3 \cdot 2^{a-1})/3 \ge 2^{a-2} T(2^{a-1}) \ge 4T(2^{a-1})$. The sum is therefore strictly positive. ∎

The constant 3 cannot be improved by any argument that uses only the soft properties in Section 5: a model with a giant ω at index 2^a and F_a "prime" has $f^*(2^{a+1})/f^*(2^a) \to 3$.

Theorem 4. *If F_a is composite ($a \ge 4$), then $f(2n) \ge \big(4 - \frac{16}{3 \cdot 2^{a-2} + 4}\big) f(n)$ for all $n \in [2^a, 2^{a+1})$. Since $F_5, \dots, F_{32}$ are composite, $f(2n) \ge 3.4\, f(n)$ for $32 \le n < 2^{33}$.*

Proof. As above, $f(2n) - 4f(n) \ge \sum_R (T(2r) - 3C(r))$; terms with $r \ne 2^a$ are ≥ 0 by Lemma 3.2. $F_a \equiv 2 \pmod 3$ is not a square, so composite means $\tau(F_a) \ge 4$, and the $r = 2^a$ term is $\ge T(2^a) - 6T(2^{a-1}) \ge -4T(2^{a-1})$ (using $T(2^a) \ge 2T(2^{a-1})$). With $t = T(2^{a-1})/f(n)$: $f(2n)/f(n) \ge 4 - 4t$, and also $f(2n) \ge T(3 \cdot 2^{a-1}) \ge 3 \cdot 2^{a-2} T(2^{a-1})$ gives $f(2n)/f(n) \ge 3 \cdot 2^{a-2} t$. The maximum of the two is minimised where they meet. An audited sharper form, $4 - \frac{4}{2^{a-2}+4}$ (giving $11/3$ on $[32, 2^{33})$), is in the scratchpad notes. ∎

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4 Good n are everywhere

Let $D(M) = \max_{m \le M} \tau(m)$ and $J(C) = \min\{J : D(2^J) - 2 > J \log_2 C\}$. Then

$J(C) = O(\log\log C \cdot \log\log\log C)$; for example $J(4) = 8$, $J(16) = 13$, $J(10^4) = 21$.

Theorem 5. *For every C and n, some $j < J(C)$ has $f(2^{j+1}n) > Cf(2^j n)$. Consequently $\{n : f(2n) > Cf(n)\}$ has lower natural density at least $1/(2^{J(C)} - 1)$.*

Proof. Take $h \le 2^J$ with $\tau(h) = D(2^J)$. For each m, the divisors me ($e \mid h$, $e > 1$) of hm do not divide m, and all but possibly one ($= 6$) carry a new primitive prime, so $T(hm) \ge 2^{D(2^J)-2} T(m)$; summing over $m \le n$ and using that f increases, $f(2^J n) \ge 2^{D(2^J)-2} f(n)$. If all J ratios were $\le C$, then $C^J \ge 2^{D(2^J)-2}$, contradicting the choice of J. For the density, group $n \le x$ by odd part m: the chain $m, 2m, \ldots$ has $\lfloor \log_2(x/m) \rfloor + 1$ terms up to x, at least a $\lfloor \cdot / J \rfloor$ fraction of them good, and $\sum_{m \text{ odd}} \#\{i \ge 0 : m 2^{iJ+J-1} \le x\} = x/(2^J - 1) + O(J \log x)$. ∎

The bound is optimal for the chain property alone: $\{n : v_2(n) \equiv J - 1 \bmod J\}$ has density exactly $1/(2^J - 1)$ and meets every chain.

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5 The barrier

Every proof attempt in this round and the previous one used only the following soft properties of the numbers $2^k - 1$: the cyclotomic factorisation and $\gcd(2^a - 1, 2^b - 1) = 2^{\gcd(a,b)} - 1$; a primitive prime for every index $d \ne 1, 6$, $\equiv 1 \bmod d$; $T(2k) = T(k)\tau(2^k + 1)$; lifting-the-exponent multiplicities; size bounds $\Phi_d(2) \approx 2^{\varphi(d)}$; Aurifeuillian splitting (two primitive pieces for $d \equiv 4 \bmod 8$, $d > 20$); the quadratic-residue rule ($q \equiv \pm 1 \bmod 8$ iff $2 \mid (q-1)/\mathrm{ord}_q 2$) and power-residue rules of bounded depth; Stewart's lower bound for the largest prime factor; Brun–Titchmarsh and averaged sieve bounds.

Theorem 5′ (models). *Replace each $\Phi_d(2)$ by a product of genuine primes $\equiv 1 \bmod d$ chosen to satisfy every property listed above, with one prime per generic index (two at Aurifeuillian indices) and, along a sparse sequence of primes ℓ_j, $W_j \ge D(3\ell_j) + O(\log \ell_j)$ primes at index ℓ_j while index $2\ell_j$ gets one prime. Then $f^*(2\ell_j)/f^*(\ell_j) \to 5$, so $\liminf < \infty$. With the spike at 2^a instead, the ratio tends to 3. With spikes spaced by a factor X, the set of n with bounded ratio has lower logarithmic density $\ge \log(X/4)/\log(2X)$.*

Proof. The multiples of ℓ_j up to $2\ell_j$ are $\ell_j, 2\ell_j$, and the spike primes divide X_k^* only for those k; so $T^*(\ell_j) = 2^{W_j}$, $T^*(2\ell_j) = 4T^*(\ell_j)$, while every other $k \le 2\ell_j$ has $\log_2 T^*(k) \le D(2\ell_j) + \sum_{i<j} W_i$, which the choice of W_j makes negligible. Each listed

property is checked index by index; the existence of the required primes in residue classes of modulus $O(d)$ follows from Siegel–Walfisz. (Full checks and the audit are in the scratchpad notes; the earlier note's Theorem 17 is the base case.) Because the soft properties also hold in models where every $\omega_d \le d^{o(1)}$, and those models have $f^*(2n)/f^*(n) \to \infty$, the soft properties are consistent with both answers. ∎

The one property such models cannot imitate is exact divisibility: a prime q divides $2^p - 1$ iff 2 is a $\frac{q-1}{p}$-th power residue mod q, a residue condition of unbounded depth. Several standard tools were tested against it and none applies (Section 6).

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6 The exact gap

Call k *K-thin* if the odd part of k has fewer than K divisors, and let $\psi(x) = \max_{m \le x} \tau(m) = 2^{(1+o(1))\log x/\log\log x}$.

Theorem 6 (exact reduction). *For all n and $K \ge 2$, $\sum_{k \le n,\ k \text{ not } K\text{-thin}} T(k) \le \frac{f(2n)-f(n)}{2^{K-1}-1}$. Hence $f(2n)/f(n) \to \infty$ iff, for every fixed K, the T-mass of K-thin $k \le n$ is $o(f(2n) - f(n))$.*

Proof. Map k to the pair (N, b) with $N = 2^b k \in (n, 2n]$, $b \ge 1$; this is a bijection onto $\{(N, b) : N \in (n, 2n], 1 \le b \le v_2(N)\}$. Iterating $T(2m) = T(m)\tau(2^m + 1)$ and $\omega(2^m + 1) \ge \tau(\text{odd part of } m) - 1$ gives $T(N) \ge 2^{b(K-1)} T(k)$ for non-thin k; sum the geometric series in b. The equivalence follows since $f(n)$ is the thin mass plus the non-thin mass. ∎

- **Affirmative follows from**: for each K, $\omega(\Phi_d(2)) \le \psi(d)(\log d)^{-2-\varepsilon}$ for all large K-thin d. Thin k have $O(K \log k)$ divisors, so their $\log_2 T$ is $o(\psi(n))$, while $f(n) \ge 2^{\psi(n)-2}$.
- **Negative requires**: infinitely many thin k with $\tau(2^k + 1)$ bounded and some $d \mid k$ having $\omega(\Phi_d(2)) \ge \psi(d)/(K \log^2 d)$. The simplest case is a prime p with $(2^p + 1)/3$ prime and $\omega(2^p - 1) \gtrsim \psi(2p)$, or a Fermat prime F_a preceded by a Fermat number F_{a-1} with $\gtrsim \psi(2^a)$ prime factors.
- **The best unconditional bound** is $\omega(2^p - 1) \le (\frac{\log 2}{2} + o(1))\, p/\log p$ (the i-th prime factor exceeds $2pi$). The needed bound is $p^{(\log 2 + o(1))/\log\log p}$. Primes below $p^{1+c/\log\log p}$ or above $\exp(p^{1-(\log 2+o(1))/\log\log p})$ are harmless; everything turns on how many prime factors of $2^p - 1$ lie in between.

Tested against the gap, each failing for a stated quantitative reason:

- **Ideal factorisation of $(2-\zeta_p)$ in $\mathbb{Q}(\zeta_p)$:** it is the Chinese-remainder decomposition of $\mathbb{Z}/(2^p-1)$ and carries no extra information.
- **Class groups, Stickelberger, Eisenstein reciprocity:** the constraints they impose are single relations or congruences, each of which a fake model can satisfy.
- **Chebotarev, even under GRH:** the residue condition lives in a field whose degree exceeds q.
- **Baker–Matveev and Yu:** lower bounds on factorisation complexity, which weaken as ω grows.
- **abc and Evertse's S-unit counts:** they constrain powerful parts or depend only on the number of primes involved, so they never cap ω.
- **Coppersmith–Howgrave-Graham–Nagaraj divisors in residue classes:** worse than trivial here.
- **Large sieve, Brun–Titchmarsh, Bourgain–Glibichuk–Konyagin:** averaged or wrong range.
- **Skolem–Mahler–Lech / LTE transfer:** any prime factor of 2^p-1 divides 2^k-1 only when $p \mid k$, so no information transfers to indices outside the spike.

A Poisson model with $\omega(\Phi_d(2)) \approx \log d$ on average makes spikes finite almost surely, so the expected answer is $f(2n)/f(n) \to \infty$. That is heuristic and is not used in any proof above.

A Data

2^k-1 is completely factored for all $k \le 300$; every factorisation multiplies back exactly and all 2188 prime factors are certified prime (FLINT, primality proofs, no probable primes). Not used in any proof.

n	$f(2n)/f(n)$	NOTE
1, 2, 4	3.000	equality in Theorem 3

8	3.741	F_3 prime
16	5.337	minimum over $16 \le n \le 150$; F_4 prime
32	15.13	F_5 composite
64	18.45	
128	154.8	

Local minima of the ratio sit at powers of 2, the configuration that Theorems 3–4 single out.

Extension to $k \le 600$: exact for $k \le 348$; for 33 indices in $[349, 599]$ (all of them d itself, with few divisors) one cofactor of 93–170 digits remains, every prime factor of which exceeds $1.4 \cdot 10^{10}$, giving rigorous intervals. $T(1..300)$ agrees with OEIS A046801. The smallest ratio for $150 \le n \le 300$ is 256.1. Records of T up to 600 are all multiples of 12 from $N = 12$ on ($12, 18, 20, 24, 36, 48, 60, 72, 84, 108, 120, 144, 168, 180, 288, 300, 360, 420, 540$); $\max_{d<600} \omega(\Phi_d(2)) = 8$ and $\max_{p<600} \omega(2^p - 1) = 8$ (at $p = 431$), nowhere near a spike.